Southern New Hampshire University

CS 499: Module Milestone Two Narrative

I chose to enhance the Grazioso Salvare Animal Shelter Dashboard that I originally developed in CS 340 using Python, Dash, and MongoDB. At the time, my focus was completing the FIX MEs to meet the functional requirements. I always thought it was one of the coolest projects I've worked on, but there were several things I wanted to improve. Most of the application was contained in a Jupyter Notebook in Codio, making it difficult to maintain or expand. I wanted to restructure the application and make it something I could run locally and include in my portfolio without relying on Codio.

I converted the notebook into a standalone Dash application and separated the filtering, validation, database operations, and authentication into different modules. I also redesigned the dashboard and added functionality for staff to create and edit animal records. These changes required input validation, database updates, and staff authentication. I wrote automated unit tests to verify that the validation, database operations, and authentication worked as intended.

These enhancements address Course Outcomes 3, 4, and 5. Outcome 3 is demonstrated through the improved application structure and separation of responsibilities. Outcome 4 is reflected in the new functionality and redesigned interface, which make the dashboard more practical for shelter staff. For Outcome 5, I incorporated security through staff authentication, input validation, and restrictions on certain database fields. Although the authentication is intended for local use, it helped me better understand the importance of considering security during development.

The enhancement was more involved than I anticipated, especially when moving everything out of Codio and getting the forms to communicate with MongoDB. I had to verify that changes actually persisted after refreshing the dashboard. My original goals remained the

same, but implementing them required more troubleshooting than expected. Writing automated tests also helped me catch problems without repeatedly testing everything manually. Overall, the experience reinforced that getting an application to work is only part of development. Making it maintainable, reliable, and secure is just as important.